

Prequential Answer-Set Calibration for Temporal Knowledge Graph Forecasting

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Abstract

Forecasts from temporal knowledge graphs can inform event monitoring and downstream decisions, yet standard ranking metrics do not indicate whether a returned answer set covers the recorded truth at a stated rate. Existing conformal methods for knowledge graphs mainly calibrate static link predictions. Can calibration use only previously revealed timestamp batches to repair reliability under temporal ordering, and what utility does that repair cost? We study this question with a leakage-controlled prequential protocol rather than claim validity under arbitrary drift. The evaluation crosses ICEWS14 and ICEWS05-15, two matched-protocol scoring families, five seeds, four score transformations, three history policies, controlled deletion of training facts, and filtered-negative sensitivity. At 30% deletion and a 0.90 target, rolling margin calibration attains 0.8988–0.9002 observed-label coverage. Relative to a fixed static margin threshold, it gains 0.0798–0.1025 coverage but adds 1158–2038 entities per set. A factorial ablation shows that history and score geometry interact: rolling Minmax saves 1091–1773 entities relative to rolling margin, with dataset-dependent coverage effects, whereas rolling Softmax overcovers and adds 2081–4991 entities. Training-fact deletion and the tested negative-sampling choice do not explain the main reliability gap. Multi-answer full-set coverage remains markedly below single-answer coverage, and a query-max calibration objective further reduces full-set coverage by 0.0011–0.0081. Recent-history calibration is therefore a useful temporal reliability diagnostic, but not a uniformly efficient or universally valid solution.

Keywords: temporal knowledge graph; link forecasting; conformal prediction; prequential evaluation; uncertainty quantification; answer sets; temporal shift

1. Introduction

Temporal knowledge graphs (TKGs) encode time-stamped relations among entities. They support forecasts of future interactions and events in settings where the cost of overlooking a plausible answer may exceed the cost of inspecting a short list. Most TKG models are nevertheless evaluated with filtered mean reciprocal rank (MRR) and Hits@K [1–4]. These metrics assess ordering. They do not answer a different operational question: does a set returned with nominal coverage $1 - \alpha$ actually contain the recorded correct entity at that rate?

Conformal prediction turns model scores into prediction sets without requiring a parametric label-probability model [5,6]. KGCP applies split conformal prediction to static knowledge graph embeddings, and CondKGCP studies predicate-conditional answer sets [7,8]. Temporal forecasting changes the information structure. Calibration precedes the test stream, score distributions may evolve, and a label observed at timestamp t must

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not influence another prediction made for the same timestamp. A fixed threshold may therefore cease to be reliable even when the ranking model itself is unchanged.

This paper asks: *can an answer-set threshold updated only from strictly past timestamp batches repair temporal reliability, and what is paid in set size and subgroup behavior?* We answer with a timestamp-batched prequential protocol. A frozen scorer first emits every subject- and object-side prediction at time t . Only then are labels from that batch released to the calibration history. Static, expanding, and recent rolling histories are crossed with four nonconformity scores so that the effect of history is not confused with the effect of score geometry.

The evidence is broader than a single-model case study but remains deliberately controlled. The main matrix covers ICEWS14 and ICEWS05-15, temporal DistMult and a matched-protocol continuous-time complex scorer, five seeds, and deletion of up to 30% of training facts. Paired sensitivity runs compare filtered and uniform training negatives. A follow-up factorial analysis reuses verified checkpoints and evaluates four scores under three history policies. In total, the primary matrix contains 90 completed seed–deletion conditions, while the factorial analysis contains 20 dataset–scorer–seed cases and 131,550 timestamp–method rows.

Three findings shape the contribution. First, a recent-history margin threshold repairs the severe undercoverage of a fixed margin threshold in all four dataset–scorer cases, but requires much larger sets. Second, the best reliability–utility trade-off is not stable across score functions or datasets: Minmax is markedly smaller on ICEWS14, whereas Softmax becomes inefficiently conservative under rolling updates. Third, marginal label coverage conceals a large multi-answer deficit, and the most direct query-max calibration attempt does not correct it. The contribution is a leakage-controlled diagnostic, a factorized empirical account of its failure modes, and an explicit boundary on what temporal conformal claims the evidence can support. It is not a new distribution-free theorem for arbitrary temporal drift.

2. Related Work

Temporal knowledge graph forecasting. TKG completion methods include temporal sequence encoders, tensor factorizations, and recurrent graph networks [1–3,9]. Simple historical baselines can remain competitive, which makes split design and temporal information control central to evaluation [10,11]. Our calibration layer consumes a frozen all-entity score vector and does not change candidate ranking. The two scorer families used below are therefore controlled protocol variants, not new forecasting architectures.

Conformal prediction for graph outputs. Probability calibration has been studied for neural classifiers and knowledge graph embeddings [12,13]. KGCP constructs answer sets for link prediction with NegScore, query-wise Minmax, and Softmax transformations [7]; CondKGCP moves from marginal toward predicate-conditional behavior [8]. APS and RAPS use ranked probability mass to trade coverage against set size in multiclass prediction [14]. These studies make score design a first-order experimental factor rather than a cosmetic implementation choice.

CFEP recently addressed conformal event-type prediction on temporal knowledge graphs with score diffusion and learned quantile weights [15]. Our estimand and protocol differ. We study entity answer sets for link forecasting, freeze the scorer, and isolate how past-only history and score geometry affect calibration. The aim is diagnostic: the analysis reports where a simple temporal update works, what it costs, and where it fails.

Calibration under non-exchangeability. Weighted conformal prediction, Adaptive Conformal Inference, and analyses of non-exchangeable data provide guarantees under specified shift or online feedback conditions [16–19]. NCPNET studies non-exchangeable

temporal graph neural networks [20]. Our rolling rule is intentionally simpler: it estimates an empirical quantile from the most recent revealed scores. Because it uses no likelihood-ratio weights and assumes no bounded drift process, exchangeable coverage cannot simply be transferred to the test stream.

3. Problem, Calibration, and Guarantee Boundary

3.1. Temporal Queries and Reliability Targets

Let $\mathcal{E}$ and $\mathcal{R}$ be closed entity and relation vocabularies. A TKG contains quadruples (s, r, o, t) . We write an object query as $x = (s, r, \square, t)$, where $\square$ denotes the entity slot to be predicted, and represent subject prediction by $(o, r^{-1}, \square, t)$. For a query x , let $A_t(x) \subseteq \mathcal{E}$ be all recorded answers at timestamp t , and let $C_t(x)$ be the predicted answer set.

The primary reliability estimand is observed-label coverage,

$$\widehat{\text{Cov}}_{\text{label}} = \frac{1}{N} \sum_{(x,y)} \mathbb{1}\{y \in C_t(x)\}.$$

It differs from full-set query coverage, $\mathbb{1}\{A_t(x) \subseteq C_t(x)\}$. We report both, together with mean set size, filtered MRR, and partial answer recall $|A_t(x) \cap C_t(x)| / |A_t(x)|$. Each quantity refers to labels recorded in the benchmark, not to every event that might be true outside the data.

3.2. Scores and History Policies

Let $f_\theta(x, y)$ be the score assigned by a frozen TKG model. Margin nonconformity is

$$a_{\text{margin}}(x, y) = \max_{z \in \mathcal{E}} f_\theta(x, z) - f_\theta(x, y), \quad C_q(x) = \{y : a(x, y) \leq q\}. \quad (1)$$

For calibration scores $a_1, \dots, a_n$ and target $1 - \alpha$, let

$$k = \min\{n, \lceil (n+1)(1-\alpha) \rceil\}, \quad q = a_{(k)}. \quad (2)$$

Static history estimates q once from the final initial-calibration interval. Expanding history uses every score revealed before the current timestamp. Rolling history uses the most recent $W = 1000$ revealed scores.

To separate the score from the history, the factorial study also evaluates the three transformations used by KGCP. For score vector f ,

$$\begin{aligned} a_{\text{Neg}}(y) &= -f_y, \\ a_{\text{Minmax}}(y) &= -\frac{f_y - \min_z f_z}{\max_z f_z - \min_z f_z}, \\ a_{\text{Softmax}}(y) &= 1 - \frac{\exp(f_y/T)}{\sum_z \exp(f_z/T)}, \end{aligned} \quad (3)$$

where zero-range Minmax vectors map to zero and $T = 1$. We reproduce these published score definitions under our common split and scorers; we do not claim to reproduce the complete KGCP training pipeline.

At timestamp t , all query scores, thresholds, and sets are stored before any label from t is released. The entire batch is then appended to the expanding and rolling histories. This ordering prevents within-timestamp leakage.

3.3. Coverage Bounds and Set Geometry

The first theorem states the familiar exchangeable benchmark against which the temporal experiment should be read. It is included to make clear which part of the usual conformal argument is lost after a chronological split [5,6].

Theorem 1 (exchangeable split-conformal reference). Condition on a frozen scorer. Suppose $a_1, \dots, a_n, a_{n+1}$ are exchangeable and $\alpha \geq 1/(n+1)$. With q from Equation (2),

$$\Pr\{a_{n+1} \leq q\} \geq 1 - \alpha. \quad (4)$$

If the scores are almost surely distinct, the coverage is at most $1 - \alpha + 1/(n+1)$.

Proof. Exchangeability makes the rank of a_{n+1} among the $n+1$ scores uniform. The event that this rank does not exceed $\lceil (n+1)(1-\alpha) \rceil$ has probability at least $1 - \alpha$. Using the calibration order statistic can only include tied test scores; without ties, the next attainable rank contributes at most $1/(n+1)$. $\square$

Temporal ordering need not satisfy the premise of Theorem 1. The next result does not restore that guarantee; it decomposes the error of an empirical threshold. Condition on all information available before time t . Let F_t be the CDF of the current observed-label score, $\hat{H}_t$ the empirical CDF of the selected past-score pool, and H_t a reference CDF for that pool. Define

$$\delta_t = \sup_u |F_t(u) - H_t(u)|, \quad \epsilon_t = \sup_u |\hat{H}_t(u) - H_t(u)|.$$

Let $\eta_t = \hat{H}_t(q_t) - \hat{H}_t(q_t^-)$ be the empirical mass tied at the selected threshold, and $\rho_{n,\alpha} = \max\{0, (k-1)/n - (1-\alpha)\}$.

Theorem 2 (deterministic temporal-mismatch decomposition). For the inclusive threshold in Equation (2),

$$1 - \alpha - \delta_t - \epsilon_t \leq F_t(q_t) \leq 1 - \alpha + \rho_{n,\alpha} + \eta_t + \delta_t + \epsilon_t, \quad (5)$$

with both bounds clipped to $[0, 1]$.

Proof. By construction, $\hat{H}_t(q_t) \geq k/n \geq 1 - \alpha$. Two applications of the sup-norm bounds yield the lower inequality. Immediately before q_t , at most $k-1$ empirical observations are strictly smaller, so $\hat{H}_t(q_t) \leq (k-1)/n + \eta_t$. Applying the same two deviations gives the upper inequality. $\square$

The decomposition identifies a trade-off. A shorter history may reduce temporal mismatch δ_t , yet increase empirical error ϵ_t or the effect of atoms η_t . These terms are not estimated as a new guarantee; the experiments measure their observable consequence through coverage and set size.

Proposition 1 (set nesting and multi-answer criterion). For any nonconformity score, $q_1 \leq q_2$ implies $C_{q_1}(x) \subseteq C_{q_2}(x)$. For margin nonconformity and $q \geq 0$, the set is nonempty. Moreover,

$$A_t(x) \subseteq C_q(x) \iff \max_{y \in A_t(x)} a(x, y) \leq q. \quad (6)$$

Proof. Prediction sets are nonconformity sublevel sets and therefore nested. Under Equation (1), a top-scoring entity has score zero. The final equivalence follows because every recorded answer is included exactly when its largest nonconformity score does not exceed the threshold. $\square$

Equation (6) motivates a query-max calibration objective. It does not imply that its empirical threshold will dominate a label-level threshold: changing from labels to unique queries changes both the sample size and the order statistic. We test that distinction directly.

Table 1. Normalized dataset statistics. Relations exclude inverse copies used internally for subject prediction.

Dataset	Entities	Relations	Facts	Timestamps
ICEWS14	7,128	230	90,730	365
ICEWS05-15	10,488	251	461,329	4,017

4. Experiments and Results

4.1. Study Design

The experiments answer four questions. **RQ1** asks whether recent-history calibration repairs static undercoverage across datasets and scorers. **RQ2** separates history policy from score geometry and measures their utility cost. **RQ3** tests whether training-fact deletion or filtered negative sampling explains the main effect. **RQ4** measures multi-answer reliability and tests the query-max objective from Proposition 1.

ICEWS14 and ICEWS05-15 are event graphs derived from the ICEWS coded event data [21]. Table 1 reports the normalized graph sizes. Unique timestamps are sorted: the earliest 60% train the scorer, the next 20% provide disjoint validation and calibration roles, and the final 20% form the test stream. The middle interval is divided chronologically into 25% scorer validation, 30% calibrator tuning, 15% selector validation, and 30% final initial calibration.

Temporal DistMult uses 256-dimensional real embeddings. The second scorer uses 128-dimensional complex embeddings and a shared continuous-time map formed by constant, linear, quadratic, sine, and cosine features. It is a matched-protocol ComplEx-style alternative, not the official TComplEx or TNTComplEx optimizer. Both scorers use the same pairwise softplus objective, Adam, learning rate 10^{-3} , weight decay 10^{-6} , batch size 1024, 128 negatives, at most 200 epochs, and validation-based early stopping.

The primary matrix uses filtered training negatives and seeds {17, 29, 43, 59, 71}. ICEWS14 uses both scorers at deletion rates {0, 0.1, 0.2, 0.3}. ICEWS05-15 uses all four rates for DistMult and rates {0, 0.3} for the continuous complex scorer. Deletion affects training facts only and retains an entity's earliest occurrence if uniform deletion would otherwise remove all of its training occurrences. Paired uniform-negative sensitivity is run for ICEWS05-15 DistMult and ICEWS14 continuous complex at deletion rates 0 and 0.3. Filtered negatives exclude positives retained in the training graph and do not consult future facts.

The factorial follow-up fixes deletion at 0.3, uses filtered negatives, and reuses the verified checkpoint for each of the four dataset-scorer cases and five seeds. It crosses the four scores in Equation (3) with static, expanding, and rolling histories. It also evaluates query-max margin under the same three histories. No test result selects a score or policy.

The target coverage is 0.90. Primary intervals use 20,000 equal-seed circular moving-block bootstrap replicates with seven-timestamp blocks; block lengths 3, 14, and 21 test sensitivity to the dependence scale. Contrasts are paired by seed and timestamp. Code, resolved configurations, derived result tables, figure scripts, and checksums are available at <https://github.com/xywang815/riskcal-tkg-w>. Raw ICEWS data are available from <https://doi.org/10.7910/DVN/28075> and the public benchmark source documented in the repository.

4.2. RQ1: Recent History Repairs Static Margin Undercoverage

Figure 1 first establishes the cross-dataset pattern. At 30% deletion, rolling margin coverage is 0.8996 and 0.8988 on ICEWS14 with DistMult and the continuous complex scorer, and 0.9002 for both scorers on ICEWS05-15. The paired rolling-minus-static gains

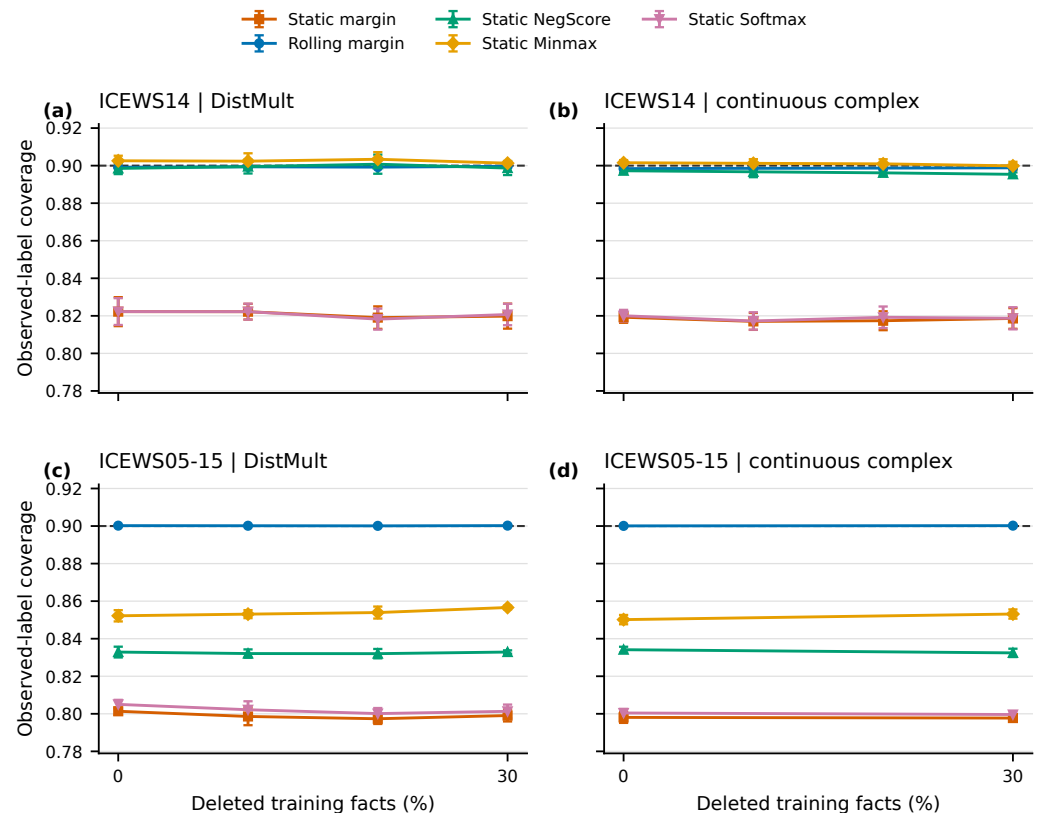


Figure 1. Observed-label coverage across deletion rates. Points are means over five seeds, error bars show one seed standard deviation, and the dashed line is the 0.90 target. ICEWS05-15 continuous complex is evaluated at deletion rates 0 and 0.3. Methods share the same scorer within each panel.

are 0.0798 (95% CI [0.0621, 0.1006]), 0.0802 ([0.0612, 0.1014]), 0.1011 ([0.0921, 0.1102]), and 0.1025 ([0.0938, 0.1111]), respectively. The corresponding filtered MRR values are 0.3103, 0.3129, 0.3269, and 0.3295; calibration changes answer-set membership but leaves this ranking unchanged.

The repair is not free. In the factorized follow-up, rolling margin increases mean set size over static margin by 1175 entities on ICEWS14 DistMult (95% CI [959, 1416]), 1158 on ICEWS14 continuous complex ([947, 1382]), 2026 on ICEWS05-15 DistMult ([1867, 2183]), and 2038 on ICEWS05-15 continuous complex ([1888, 2182]). The absolute target-error reduction is positive for every case and every tested block length, as is the set-size increase. The reliable part of the conclusion is therefore a paired movement toward the target accompanied by a substantial loss of efficiency.

4.3. RQ2: History and Score Geometry Interact

Figure 2 separates the two components that the initial study coupled. Margin and Softmax change sharply as the history moves from static to expanding to rolling. NegScore and Minmax are comparatively stable on ICEWS14, where their static thresholds already lie near 0.90. On ICEWS05-15, all four scores respond to the history choice, but they occupy different regions of the coverage–size plane.

Within rolling history, Minmax changes ICEWS14 coverage by only 0.0001 and 0.0005 relative to margin, with both 95% intervals crossing zero, while reducing mean set size by 1773 and 1650 entities. On ICEWS05-15 it reduces size by 1187 and 1091 entities, but its coverage changes of -0.0014 and -0.0011 are small and sensitive to block length. NegScore shows the same dataset dependence: it is smaller without a resolved coverage change on ICEWS14, but slightly less reliable and not consistently smaller on ICEWS05-15.

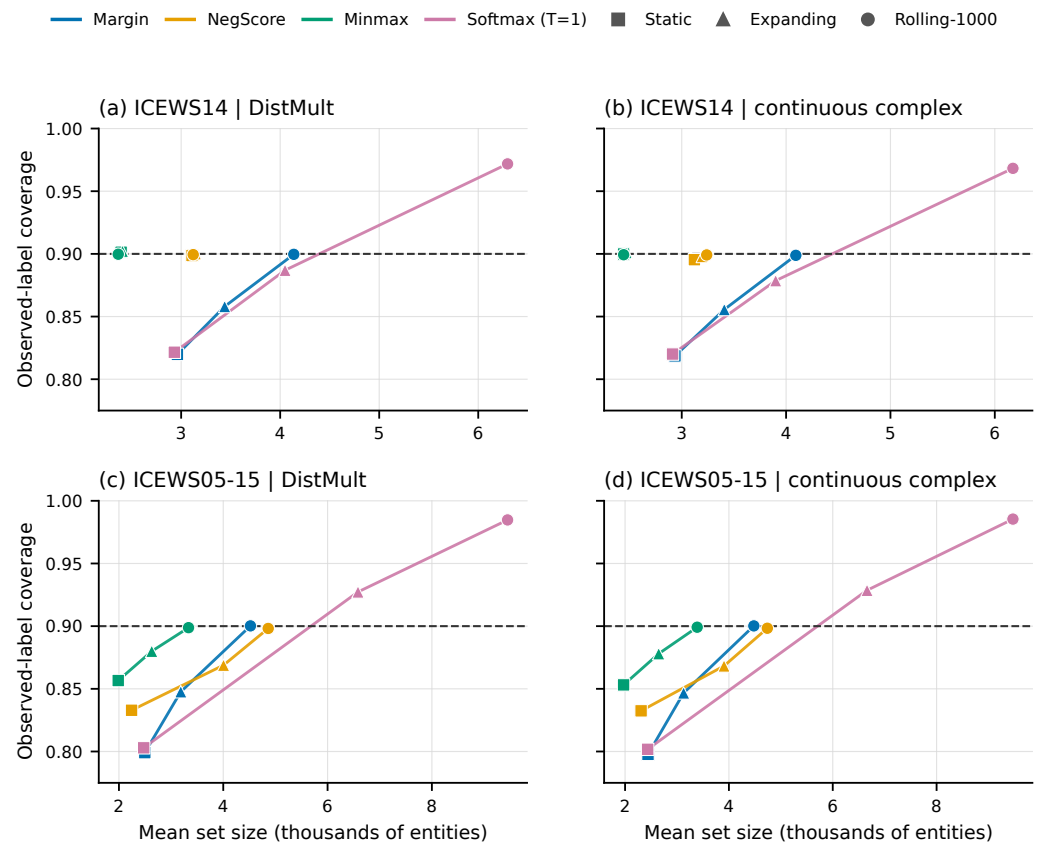


Figure 2. Factorial ablation of nonconformity score and calibration history at 30% deletion. Each colored path connects static (square), expanding (triangle), and rolling-1000 (circle) histories for one score. The dashed line is the 0.90 target. Values are equal-seed means; paired uncertainty is reported in the text.

Softmax illustrates why coverage alone is insufficient. Under rolling history, it raises coverage over margin by 0.069–0.085, yet adds 2081–4991 entities. The resulting coverage of 0.9683–0.9854 is overconservative and leaves little screening value on the larger graph. No score function dominates across all cases; score geometry and temporal history must be assessed jointly.

A separate ICEWS14 DistMult shortlist diagnostic reached the same caution from a different construction. A validation-selected RAPS-style rolling rule used softmax temperature $T = 1$, selection tolerance 0.02, and a fixed 21-candidate grid comprising APS plus $k \in \{50, 100, 250, 500, 1000\}$ and $\lambda \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$. All 20 seed-deletion conditions selected $k = 50$ and $\lambda = 10^{-4}$. At 30% deletion it reduced mean set size from 4069.4 for rolling margin to 3319.0 while changing observed-label coverage from 0.8998 to 0.8987; full-set query coverage fell from 0.9020 to 0.8874. This is an efficiency diagnostic under one scorer, not evidence of a general improvement.

4.4. RQ3: The Tested Data Interventions Do Not Explain the Gap

Figure 3 distinguishes the history effect from two proposed mechanisms. For 30% minus 0% deletion, every static-margin and rolling-margin coverage interval crosses zero. Static changes range from -0.0024 to -0.0004, and rolling changes range from 0.0000 to 0.0005. Training-fact deletion is thus a scorer stress test, but the experiment does not support it as the cause of static calibration failure.

Filtered-minus-uniform rolling effects also cross zero in all four paired conditions. A small positive static-margin effect appears only for ICEWS05-15 DistMult without deletion

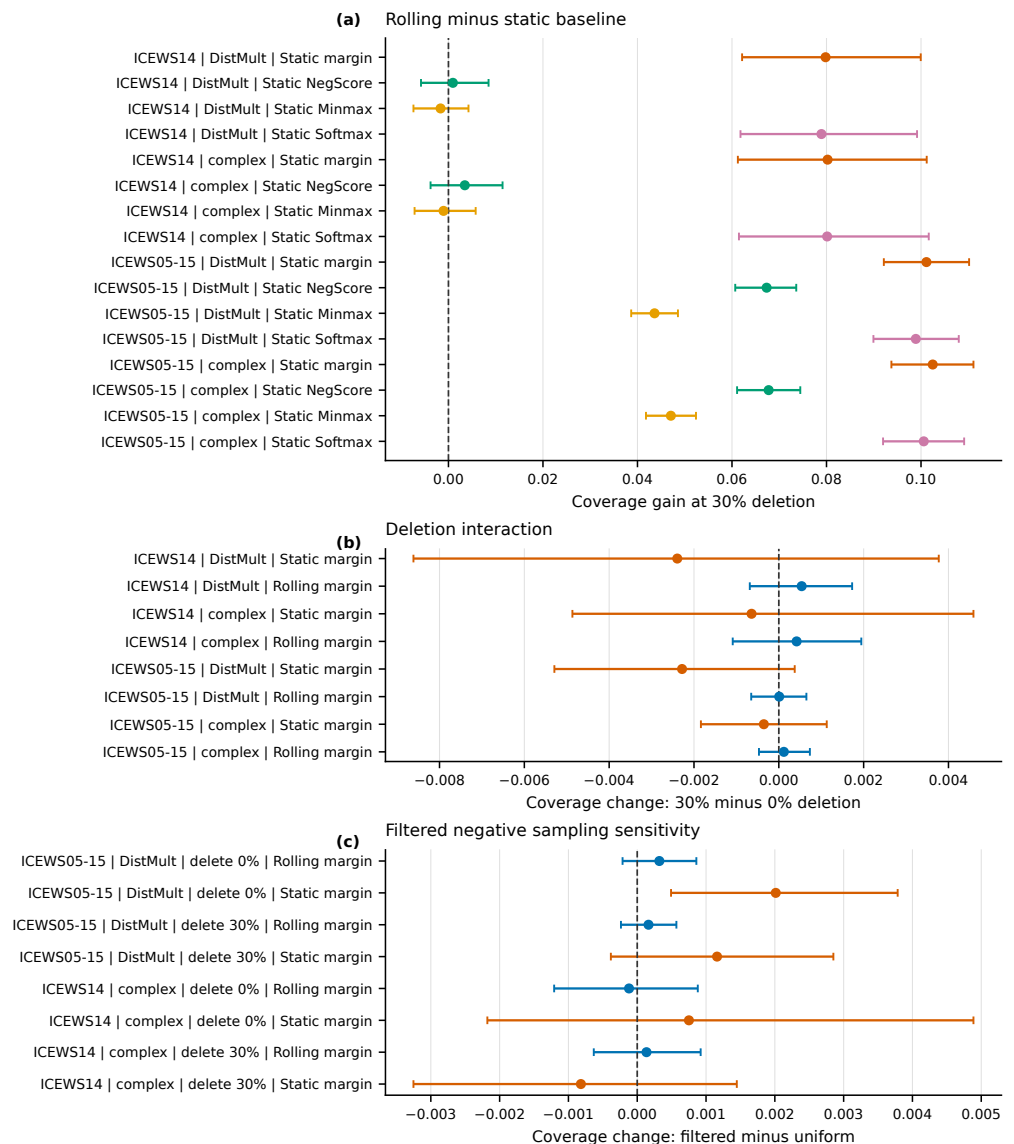


Figure 3. Paired timestamp-block bootstrap diagnostics (20,000 replicates; seven-timestamp blocks). (a) Rolling-minus-static-baseline coverage at 30% deletion. (b) Coverage change from deleting 30% rather than 0% of training facts. (c) Coverage change from filtered rather than uniform training negatives. Intervals crossing zero are inconclusive.

(0.0020 [0.0005, 0.0038]); the remaining static intervals are inconclusive. The main history result is not attributable to the tested negative-sampling choice.

4.5. RQ4: Label Coverage Conceals Multi-Answer Failure

In the factorial follow-up, under rolling label-margin calibration, multi-answer full-set coverage is lower than single-answer coverage in every case: the mean gaps are -0.2773 and -0.3015 on ICEWS14, and -0.2059 and -0.2137 on ICEWS05-15, for DistMult and continuous complex respectively. Marginal label coverage near 0.90 therefore does not imply that a set contains all recorded answers to a query.

Proposition 1 suggests calibrating the maximum nonconformity over each recorded answer set. Empirically, that direct query-max rule does not solve the problem (Figure 4). Relative to label-level rolling margin, its full-set coverage changes by -0.0081 (95% CI [-0.0107, -0.0058]) and -0.0070 ([-0.0093, -0.0049]) on ICEWS14. The ICEWS05-15 changes are -0.0011 ([-0.0022, 0.0001]) and -0.0014 ([-0.0024, -0.0003]). Sets shrink by 39–141 entities,

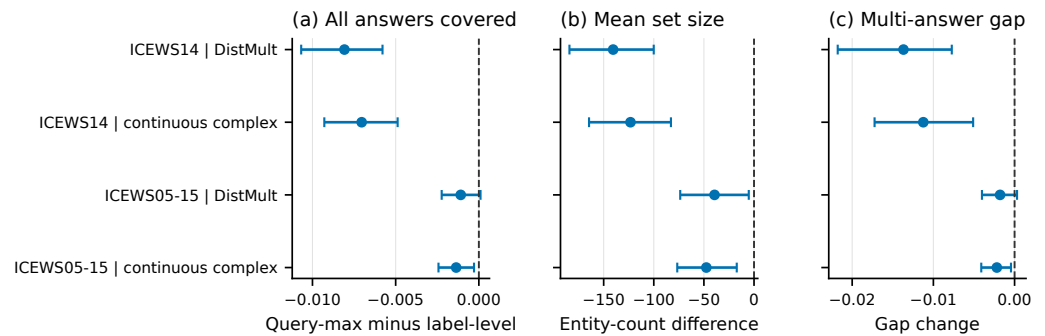


Figure 4. Query-max rolling margin minus label-level rolling margin at 30% deletion. Points are paired effects and bars are 95% circular moving-block bootstrap intervals. Panels report full-set query coverage, mean set size, and the multi-answer-minus-single-answer coverage gap. Negative values in panels (a) and (c) indicate worse query-level reliability.

but the multi-answer gap becomes more negative. Reducing the number of calibration units from labels to unique queries changes the empirical quantile enough to offset the apparently conservative maximum score.

5. Discussion and Conclusions

The central result is narrower, and more useful, than a universal method claim. When a margin threshold is fixed early, coverage later in the chronological test stream falls to roughly 0.80–0.82. Re-estimating that threshold from the most recent strictly past scores restores aggregate observed-label coverage near 0.90 in both ICEWS datasets and both controlled scorer families. Theorem 2 offers the corresponding diagnostic interpretation: recent history can reduce the discrepancy between the past-score reference and the current score distribution. The data do not identify that discrepancy term directly, and no coverage guarantee under arbitrary drift follows.

The factorial ablation changes how this result should be used. Rolling updates are decisive for margin, but score transformation can matter as much as history. Minmax is far more efficient on ICEWS14, while its ICEWS05-15 coverage effect is mixed. Rolling Softmax obtains higher coverage by returning most of the entity vocabulary. A deployment rule should therefore be chosen from a joint reliability–utility frontier, not from coverage or mean set size alone.

Two negative findings further limit the explanation. Removing up to 30% of training facts does not measurably worsen static coverage, and filtered training negatives have little effect in the paired sensitivity study. Temporal mismatch, rather than the tested graph corruption, is the defensible motivation for updating the threshold. The multi-answer analysis also exposes a different estimand failure: label-marginal calibration may be adequate while the event that all answers are present remains unreliable. A simple query-max score is not enough because its calibration sample and quantile change at the same time.

The study has four main boundaries. ICEWS14 and ICEWS05-15 are both political event graphs, so they do not establish domain-wide generality. The second scorer is a matched-protocol continuous complex model rather than an official TComplex or TNTComplex reproduction. That distinction is necessary because the published TKBC preprocessing places the same discrete time support in training, validation, and test, whereas the present protocol requires unseen future timestamps; a model with a free embedding for every future time would either be untrained there or leak temporal information. Likewise, the KGCP comparisons reuse published score transformations, not the authors' full training

pipeline. Finally, five seeds limit inference over optimization randomness, and a moving-block bootstrap only approximates the dependence structure.

These limits point to concrete next tests. Temporal answer-set calibration should be evaluated with inductive or extrapolative TKG backbones on non-ICEWS domains. Structured calibration objectives should preserve query-level sample size or explicitly model answer multiplicity. Predicate-conditional temporal rules may reduce the large relation and direction heterogeneity that marginal coverage averages away. For large entity vocabularies, any reliability gain must also be paired with candidate-generation or retrieval methods that bound the cost of all-entity scoring.

In conclusion, recent-history prequential calibration repairs a reproducible aggregate undercoverage pattern, but it does so with thousands of additional candidates and leaves multi-answer reliability unresolved. The practical contribution is a protocol for auditing that trade-off without current-timestamp leakage, together with evidence showing which apparently plausible explanations and remedies do not survive controlled tests.

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References

1. García-Durán, A.; Dumančić, S.; Niepert, M. Learning Sequence Encoders for Temporal Knowledge Graph Completion. In Proceedings of the Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 4816–4821. <https://doi.org/10.18653/v1/D18-1516>.
2. Li, Z.; Jin, X.; Li, W.; Guan, S.; Guo, J.; Shen, H.; Wang, Y.; Cheng, X. Temporal Knowledge Graph Reasoning Based on Evolutional Representation Learning. In Proceedings of the Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021, pp. 408–417. <https://doi.org/10.1145/3404835.3462963>.
3. Li, Y.; Sun, S.; Zhao, J. TiRGN: Time-Guided Recurrent Graph Network with Local-Global Historical Patterns for Temporal Knowledge Graph Reasoning. In Proceedings of the Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, 2022, pp. 2152–2158. <https://doi.org/10.24963/ijcai.2022/299>.
4. Cai, B.; Xiang, Y.; Gao, L.; Zhang, H.; Li, Y.; Li, J. Temporal Knowledge Graph Completion: A Survey. In Proceedings of the Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, 2023, pp. 6545–6553. <https://doi.org/10.24963/ijcai.2023/734>.

5. Vovk, V.; Gammerman, A.; Shafer, G. *Algorithmic Learning in a Random World*; Springer, 2005. <https://doi.org/10.1007/b106715>. 343
6. Angelopoulos, A.N.; Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *Foundations and Trends in Machine Learning* **2023**, *16*, 494–591. <https://doi.org/10.1561/2200000101>. 344
7. Zhu, Y.; Potyka, N.; Pan, J.; Xiong, B.; He, Y.; Kharlamov, E.; Staab, S. Conformalized Answer Set Prediction for Knowledge Graph Embedding. In Proceedings of the Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, 2025, pp. 731–750. <https://doi.org/10.18653/v1/2025.naacl-long.32>. 345
8. Zhu, Y.; Hernández, D.; He, Y.; Ding, Z.; Xiong, B.; Kharlamov, E.; Staab, S. Predicate-Conditional Conformalized Answer Sets for Knowledge Graph Embeddings. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 4145–4167. <https://doi.org/10.18653/v1/2025.findings-acl.215>. 346
9. Lacroix, T.; Obozinski, G.; Usunier, N. Tensor Decompositions for Temporal Knowledge Base Completion. In Proceedings of the International Conference on Learning Representations, 2020. 347
10. Gastinger, J.; Meilicke, C.; Errica, F.; Szttyler, T.; Schülke, A.; Stuckenschmidt, H. History Repeats Itself: A Baseline for Temporal Knowledge Graph Forecasting. In Proceedings of the Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, 2024, pp. 4016–4024. <https://doi.org/10.24963/ijcai.2024/444>. 348
11. Gastinger, J.; Huang, S.; Galkin, M.; Loghmani, E.; Parviz, A.; Poursafaei, F.; Danovitch, J.; Rossi, E.; Koutis, I.; Stuckenschmidt, H.; et al. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. In Proceedings of the Advances in Neural Information Processing Systems, 2024, Vol. 37. Datasets and Benchmarks Track, <https://doi.org/10.52202/079017-4450>. 349
12. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On Calibration of Modern Neural Networks. In Proceedings of the Proceedings of the 34th International Conference on Machine Learning, 2017, pp. 1321–1330. 350
13. Safavi, T.; Koutra, D.; Meij, E. Evaluating the Calibration of Knowledge Graph Embeddings for Trustworthy Link Prediction. In Proceedings of the Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020, pp. 8308–8321. <https://doi.org/10.18653/v1/2020.emnlp-main.667>. 351
14. Angelopoulos, A.N.; Bates, S.; Malik, J.; Jordan, M.I. Uncertainty Sets for Image Classifiers Using Conformal Prediction. In Proceedings of the International Conference on Learning Representations, 2021. 352
15. Hu, C.; Cao, C.; Yuan, F.; Guo, D.; Xu, P.; Liu, Y.; Liu, Y. Conformal Event Prediction with Temporal Knowledge Graph. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2026, San Diego, California, United States, 2026; pp. 5233–5248. <https://doi.org/10.18653/v1/2026.findings-acl.258>. 353
16. Tibshirani, R.J.; Barber, R.F.; Candès, E.J.; Ramdas, A. Conformal Prediction Under Covariate Shift. In Proceedings of the Advances in Neural Information Processing Systems, 2019, Vol. 32. 354
17. Gibbs, I.; Candès, E. Adaptive Conformal Inference Under Distribution Shift. In Proceedings of the Advances in Neural Information Processing Systems, 2021, Vol. 34. 355
18. Barber, R.F.; Candès, E.J.; Ramdas, A.; Tibshirani, R.J. Conformal Prediction Beyond Exchangeability. *The Annals of Statistics* **2023**, *51*, 816–845. <https://doi.org/10.1214/23-AOS2276>. 356
19. Farinhas, A.; Zerva, C.; Ulmer, D.; Martins, A.F.T. Non-Exchangeable Conformal Risk Control. In Proceedings of the International Conference on Learning Representations, 2024. 357
20. Wang, T.; Kang, J.; Yan, Y.; Kulkarni, A.; Zhou, D. Non-exchangeable Conformal Prediction for Temporal Graph Neural Networks. In Proceedings of the Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2, 2025, pp. 3031–3042. <https://doi.org/10.1145/3711896.3737064>. 358
21. Boschee, E.; Lautenschlager, J.; O'Brien, S.; Shellman, S.; Starz, J.; Ward, M. ICEWS Coded Event Data, 2015. <https://doi.org/10.7910/DVN/28075>. 359